

Advanced Optimization

Chapter 3 — Line Search Methods

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 3

The line search iteration

$$x_{k+1} = x_k + \alpha_k p_k, \quad (3.1)$$

p_k a search direction, $\alpha_k > 0$ the **step length**. Success depends on both.

Most methods require p_k to be a **descent direction**, $p_k^T \nabla f_k < 0$, and most have the form

$$p_k = -B_k^{-1} \nabla f_k, \quad (3.2)$$

with B_k symmetric nonsingular: $B_k = I$ (steepest descent), $B_k = \nabla^2 f_k$ (Newton), or a quasi-Newton approximation. If $B_k \succ 0$ then $p_k^T \nabla f_k = -\nabla f_k^T B_k^{-1} \nabla f_k < 0$: descent is automatic.

This chapter answers

- ▶ How to choose α_k so that convergence is guaranteed from *remote* starting points (§3.1–3.2)
- ▶ How fast steepest descent, Newton, and quasi-Newton then converge (§3.3)
- ▶ What to do when $\nabla^2 f_k$ is not positive definite and p_k^N is not descent (§3.4)
- ▶ How a step-length routine is actually built (§3.5)

The trade-off

We want α_k to give substantial reduction in f — without spending much time choosing it.

The ideal choice is the global minimizer of the one-dimensional function

$$\phi(\alpha) = f(x_k + \alpha p_k), \quad \alpha > 0, \quad (3.3)$$

but this is too expensive: even a *local* minimizer to moderate precision costs too many evaluations of f (and perhaps ∇f).

Inexact line search

Try a sequence of candidate α 's; accept one as soon as it satisfies certain *termination conditions*. Two phases:

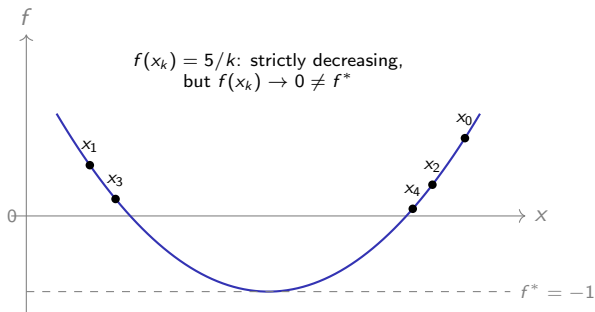
- ▶ **bracketing** — find an interval containing desirable step lengths;
- ▶ **selection** — bisection or interpolation inside it.

The conditions are the subject of §3.1; the mechanics of §3.5.

The key discovery: **an effective step length need not be anywhere near a minimizer of ϕ .**

Why “just decrease f ” is not enough (Figure 3.2)

The obvious condition, $f(x_k + \alpha_k p_k) < f(x_k)$, does not give convergence.

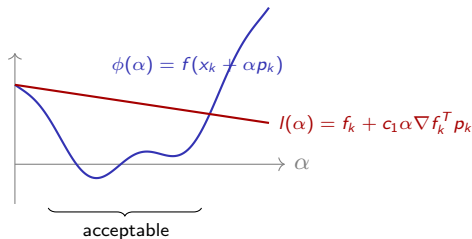


Each step decreases f , yet the iterates never reach the minimizer of this convex function. The reductions are *insufficient*. We need a **sufficient decrease** condition.

Sufficient decrease: the Armijo condition (Figure 3.3)

$$f(x_k + \alpha p_k) \leq f(x_k) + c_1 \alpha \nabla f_k^T p_k, \quad c_1 \in (0, 1). \quad (3.4)$$

The decrease must be proportional to both α and the directional derivative $\nabla f_k^T p_k$.



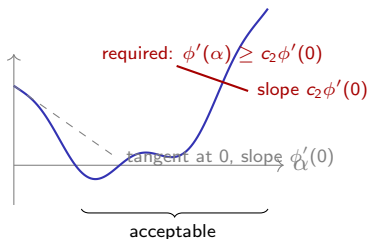
$l(\alpha)$ has slope $c_1 \nabla f_k^T p_k$ — negative, but since $c_1 < 1$ it lies *above* ϕ for small α . In practice $c_1 = 10^{-4}$: this asks for little more than descent.

Problem: it is satisfied by *every* sufficiently small α . It rules out steps that are too long, not steps that are too short.

Ruling out short steps: the curvature condition (Figure 3.4)

$$\nabla f(x_k + \alpha_k p_k)^T p_k \geq c_2 \nabla f_k^T p_k, \quad c_2 \in (c_1, 1). \quad (3.5)$$

The left side is $\phi'(\alpha_k)$: the slope at α_k must be at least c_2 times the initial slope $\phi'(0)$, i.e. *not too negative*.



Intuition. If $\phi'(\alpha)$ is still strongly negative, we can reduce f much more by going further — so don't stop. If it is only slightly negative, or positive, little more is available in this direction — stop.

Typical c_2 : **0.9** for Newton / quasi-Newton, **0.1** for nonlinear CG.

The Wolfe conditions (Figure 3.5)

Wolfe conditions

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c_1 \alpha_k \nabla f_k^T p_k, \quad (3.6a)$$

$$\nabla f(x_k + \alpha_k p_k)^T p_k \geq c_2 \nabla f_k^T p_k, \quad (3.6b)$$

with $0 < c_1 < c_2 < 1$.

A step may satisfy (3.6) without being close to a minimizer of ϕ — the curvature condition permits $\phi'(\alpha_k)$ to be arbitrarily *positive*.

Strong Wolfe conditions

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c_1 \alpha_k \nabla f_k^T p_k, \quad (3.7a)$$

$$|\nabla f(x_k + \alpha_k p_k)^T p_k| \leq c_2 |\nabla f_k^T p_k|. \quad (3.7b)$$

Now $\phi'(\alpha_k)$ may not be too positive either: α_k lies in a broad neighborhood of a stationary point of ϕ .

Wolfe conditions are scale-invariant (multiply f by a constant, affine change of variables). They are **essential for quasi-Newton methods** (Ch. 6): (3.6b) is exactly what makes $s_k^T y_k > 0$.

Do Wolfe step lengths exist? (Lemma 3.1)

Let f be continuously differentiable, p_k a descent direction, and f bounded below along the ray $\{x_k + \alpha p_k : \alpha > 0\}$. Then for any $0 < c_1 < c_2 < 1$ there exist *intervals* of step lengths satisfying (3.6) and (3.7).

Why (the argument is three lines and worth seeing):

- 1 The line $l(\alpha)$ is unbounded below; ϕ is bounded below; so they intersect. Let α' be the smallest intersection: $\phi(\alpha') = l(\alpha')$. Sufficient decrease holds on all of $(0, \alpha']$.
- 2 Mean value theorem on $[0, \alpha']$: some $\alpha'' \in (0, \alpha')$ has $\phi'(\alpha'') = \frac{\phi(\alpha') - \phi(0)}{\alpha'} = c_1 \nabla f_k^T p_k$.
- 3 Since $c_1 < c_2$ and $\nabla f_k^T p_k < 0$: $\phi'(\alpha'') = c_1 \nabla f_k^T p_k > c_2 \nabla f_k^T p_k$. So α'' satisfies both conditions, strictly — hence a whole interval around it does. And $\phi'(\alpha'') < 0$, so strong Wolfe holds there too.

Note where each hypothesis was used: bounded below (step 1), $c_1 < c_2$ (step 3), and Taylor in mean-value form (step 2) — Chapter 2's tool, already at work.

The Goldstein conditions (Figure 3.6)

$$f(x_k) + (1 - c)\alpha_k \nabla f_k^T p_k \leq f(x_k + \alpha_k p_k) \leq f(x_k) + c \alpha_k \nabla f_k^T p_k, \quad 0 < c < \frac{1}{2}. \quad (3.11)$$

- ▶ Right inequality: sufficient decrease (3.4), with $c_1 = c$.
- ▶ Left inequality: a *second line*, steeper, that ϕ must stay *above* — controls α from below *without using* ϕ' .
- ▶ Uses only function values at the trial points.

Drawback

The left inequality may *exclude all minimizers* of ϕ . Convergence theory is similar to Wolfe, but Goldstein is poorly suited to quasi-Newton methods that need to keep $B_k \succ 0$ — nothing here guarantees $s_k^T y_k > 0$.

Used in Newton-type methods; rarely elsewhere.

Backtracking (Algorithm 3.1): sufficient decrease alone can suffice

Algorithm 3.1 — Backtracking Line Search

Choose $\bar{\alpha} > 0$, $\rho \in (0, 1)$, $c \in (0, 1)$; set $\alpha \leftarrow \bar{\alpha}$.

repeat until $f(x_k + \alpha p_k) \leq f(x_k) + c \alpha \nabla f_k^T p_k$: $\alpha \leftarrow \rho \alpha$.

Terminate with $\alpha_k = \alpha$.

Why the curvature condition can be dropped. Either $\alpha_k = \bar{\alpha}$, or α_k is within a factor ρ of a value α_k/ρ that was *rejected for being too long*. Either way, α_k is not too short. The search order does the job that (3.6b) did.

- ▶ $\bar{\alpha} = 1$ for Newton and quasi-Newton; other values for steepest descent, CG.
- ▶ Terminates in finitely many trials (sufficient decrease holds for small α).
- ▶ In practice ρ varies per trial via safeguarded interpolation (§3.5), with $\rho \in [\rho_{\text{lo}}, \rho_{\text{hi}}]$ fixed.

Well suited to Newton; less so to quasi-Newton and CG — for those, the curvature condition is not a luxury.

Global convergence needs good directions too

Good step lengths are not enough; p_k must not drift toward orthogonality with ∇f_k . Measure this by the angle θ_k between p_k and $-\nabla f_k$:

$$\cos \theta_k = \frac{-\nabla f_k^T p_k}{\|\nabla f_k\| \|p_k\|}. \quad (3.12)$$

Theorem 3.2 (Zoutendijk)

Consider (3.1) with p_k a descent direction and α_k satisfying the Wolfe conditions (3.6). Suppose f is bounded below on $\mathbb{R}^n$, continuously differentiable on an open set $\mathcal{N}$ containing the level set $\mathcal{L} = \{x : f(x) \leq f(x_0)\}$, and ∇f is Lipschitz on $\mathcal{N}$: $\|\nabla f(x) - \nabla f(\tilde{x})\| \leq L\|x - \tilde{x}\|$ (3.13). Then

$$\sum_{k \geq 0} \cos^2 \theta_k \|\nabla f_k\|^2 < \infty. \quad (3.14)$$

Looks technical. Its power is that it *quantifies* what a Wolfe step buys per iteration, and thereby gives global convergence for a whole class of methods at once.

Zoutendijk: the proof is the insight

Step 1 — curvature condition gives a lower bound on α_k . From (3.6b):

$(\nabla f_{k+1} - \nabla f_k)^T p_k \geq (c_2 - 1) \nabla f_k^T p_k$. From Lipschitz: $(\nabla f_{k+1} - \nabla f_k)^T p_k \leq \alpha_k L \|p_k\|^2$. Together:

$$\alpha_k \geq \frac{c_2 - 1}{L} \cdot \frac{\nabla f_k^T p_k}{\|p_k\|^2} > 0.$$

Step 2 — sufficient decrease turns that into a per-step reduction. Substitute into (3.6a):

$$f_{k+1} \leq f_k - c_1 \frac{1 - c_2}{L} \frac{(\nabla f_k^T p_k)^2}{\|p_k\|^2} = f_k - c \cos^2 \theta_k \|\nabla f_k\|^2, \quad c = \frac{c_1(1 - c_2)}{L}.$$

Step 3 — sum, and use bounded below. $f_{k+1} \leq f_0 - c \sum_{j \leq k} \cos^2 \theta_j \|\nabla f_j\|^2$ (3.15). The left side is bounded below, so the sum converges. $\square$

What to take from it

Each Wolfe condition has a job: (3.6b) keeps α_k from collapsing; (3.6a) converts a non-collapsed step into guaranteed decrease. **Drop either one and the argument fails at exactly that step.**

Consequences of Zoutendijk

(3.14) implies $\cos^2 \theta_k \|\nabla f_k\|^2 \rightarrow 0$ (3.16). So:

- ▶ If the angle is **bounded away from** 90° , $\cos \theta_k \geq \delta > 0$ (3.17), then

$$\lim_{k \rightarrow \infty} \|\nabla f_k\| = 0. \quad (3.18)$$

In particular **steepest descent** ($\cos \theta_k = 1$) with Wolfe or Goldstein steps is globally convergent.

- ▶ **Newton-like methods** $p_k = -B_k^{-1} \nabla f_k$ with $B_k \succ 0$ and *uniformly bounded condition number* $\|B_k\| \|B_k^{-1}\| \leq M$: then $\cos \theta_k \geq 1/M$ (3.19, Exercise 3.5), so (3.18) holds again.
- ▶ For some methods (**CG**) only the weaker $\liminf_k \|\nabla f_k\| = 0$ (3.21) is available — a subsequence. Proof by contradiction: if $\|\nabla f_k\| \geq \gamma$ then $\cos \theta_k \rightarrow 0$; so it suffices to show a subsequence of $\cos \theta_k$ stays away from 0.
- ▶ A cheap trick: any method that decreases f each step and takes a steepest-descent step *every m -th iteration* satisfies (3.21).

What “globally convergent” means here

(3.18) — attracted to *stationary points*. Not “converges to a minimizer”, and certainly not “converges to the global minimizer”. Strengthening this requires negative curvature information — see Ch. 4.

Global convergence is easy; fast convergence is the problem

We could enforce (3.17) by an *angle test*: compute $\cos \theta_k$ and bend p_k toward $-\nabla f_k$ whenever it falls below some δ . Global convergence follows.

Why nobody does this

- ▶ On ill-conditioned problems the *right* direction may be nearly orthogonal to the gradient; an angle test would reject exactly the steps that converge fast.
- ▶ Angle tests destroy the invariance properties of quasi-Newton methods.

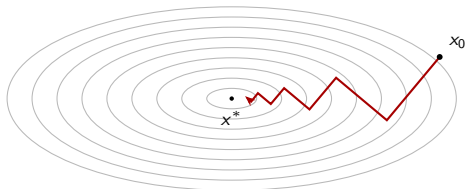
The tension of this chapter: steepest descent is the quintessential globally convergent method and *slow*; pure Newton is fast near a solution and *may not even be descent* away from it. The design problem is to get both.

We now quantify “slow” and “fast”.

Steepest descent on a quadratic (Figure 3.7)

Ideal case: $f(x) = \frac{1}{2}x^T Qx - b^T x$ (3.24), $Q \succ 0$, exact line search. $\nabla f(x) = Qx - b$, x^* solves $Qx = b$. Minimizing $f(x_k - \alpha \nabla f_k)$ over α :

$$\alpha_k = \frac{\nabla f_k^T \nabla f_k}{\nabla f_k^T Q \nabla f_k}, \quad x_{k+1} = x_k - \alpha_k \nabla f_k. \quad (3.25-3.26)$$



Contours are ellipsoids with axes along eigenvectors of Q ; the iterates **zigzag**. Each step is orthogonal to the previous one (exact line search), so progress along the long axis is slow.

How slow? (Theorems 3.3, 3.4)

Weighted norm $\|x\|_Q^2 = x^T Q x$; using $Qx^* = b$ one checks $\frac{1}{2}\|x - x^*\|_Q^2 = f(x) - f(x^*)$ (3.27), so this norm measures the objective gap.

Theorem 3.3

Steepest descent with exact line search on (3.24) satisfies

$$\|x_{k+1} - x^*\|_Q^2 \leq \left(\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1} \right)^2 \|x_k - x^*\|_Q^2, \quad (3.29)$$

$0 < \lambda_1 \leq \dots \leq \lambda_n$ the eigenvalues of Q .

- ▶ **Linear** rate; constant $\rightarrow 1$ as the condition number $\kappa = \lambda_n/\lambda_1$ grows.
- ▶ All eigenvalues equal: convergence in one step (circular contours).
- ▶ $\kappa = 800$: after 1000 iterations the objective gap is still about 8% of its initial value.

Theorem 3.4 extends this to general f : if exact-line-search steepest descent converges to x^* with $\nabla^2 f(x^*) \succ 0$, then $f(x_{k+1}) - f(x^*) \leq r^2[f(x_k) - f(x^*)]$ for any $r \in (\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}, 1)$ and all large k . Inexact line search does not improve this.

Newton's method: local quadratic convergence (Theorem 3.5)

$p_k^N = -\nabla^2 f_k^{-1} \nabla f_k$ (3.30). Here only the *local* behavior; what to do far away is §3.4 and Ch. 4.

Theorem 3.5

Let f be twice differentiable with $\nabla^2 f$ Lipschitz near a solution x^* satisfying the sufficient conditions (Thm 2.4). For $x_{k+1} = x_k + p_k^N$: (i) if x_0 is close enough to x^* , $x_k \rightarrow x^*$; (ii) the rate is quadratic; (iii) $\|\nabla f_k\| \rightarrow 0$ quadratically.

Shape of the argument. Using $\nabla f(x^*) = 0$ and Taylor's integral form (2.5),

$$x_k + p_k^N - x^* = \nabla^2 f_k^{-1} \int_0^1 [\nabla^2 f_k - \nabla^2 f(x_k + t(x^* - x_k))] (x_k - x^*) dt,$$

and the Lipschitz constant L bounds the bracket by $L t \|x_k - x^*\|$, giving

$$\|x_k + p_k^N - x^*\| \leq L \|\nabla^2 f(x^*)^{-1}\| \|x_k - x^*\|^2 = \tilde{L} \|x_k - x^*\|^2. \quad (3.33)$$

Choose $\|x_0 - x^*\| \leq \min(r, 1/2\tilde{L})$ and induct.

Note the hypotheses: $\nabla^2 f(x^*) \succ 0$, Lipschitz Hessian, *and* $\alpha_k = 1$. Near x^* the Wolfe conditions accept $\alpha_k = 1$ (next slide), so a Newton line search attains this rate.

Quasi-Newton: when is the unit step good enough? (Theorems 3.6, 3.7)

Theorem 3.6

Let $f \in C^2$, p_k descent, α_k Wolfe with $c_1 \leq \frac{1}{2}$. If $x_k \rightarrow x^*$ with $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$, and

$$\lim_{k \rightarrow \infty} \frac{\|\nabla f_k + \nabla^2 f_k p_k\|}{\|p_k\|} = 0, \quad (3.35)$$

then (i) $\alpha_k = 1$ is admissible for all large k ; (ii) if $\alpha_k = 1$ is taken, convergence is superlinear.

Why $c_1 \leq \frac{1}{2}$: with $c_1 > \frac{1}{2}$ the sufficient-decrease line excludes the minimizer of a quadratic, so the unit step can be rejected forever.

For $p_k = -B_k^{-1} \nabla f_k$, (3.35) is equivalent to the **Dennis–Moré condition**

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - \nabla^2 f(x^*))p_k\|}{\|p_k\|} = 0. \quad (3.36)$$

Theorem 3.7: with $\alpha_k \equiv 1$, superlinear convergence holds *if and only if* (3.36).

The surprising and delightful part

B_k need *not* converge to $\nabla^2 f(x^*)$. It only has to become accurate *along the search directions*. Chapter 6 shows BFGS achieves this.

The implementation detail that is not a detail

Always try $\alpha = 1$ first.

- ▶ Theorem 3.5 needs $\alpha_k = 1$ for quadratic convergence of Newton.
- ▶ Theorem 3.6 needs $\alpha_k = 1$ for superlinear convergence of quasi-Newton.
- ▶ Both theorems say the unit step *will* be accepted eventually — but only if it is *tried*. A line search that starts from last iteration's α_{k-1} , or from some scaled guess, can silently throw the fast rate away.

For methods whose directions are not naturally scaled (steepest descent, CG) the initial trial must be chosen from problem data instead — §3.5.

This is the kind of thing you will see in solver source code and log files and wonder why. Now you know.

The problem, and the generic fix

Away from x^* , $\nabla^2 f_k$ may be indefinite, and then $\nabla^2 f_k p_k^N = -\nabla f_k$ (3.38) may not give a descent direction.

Algorithm 3.2 — Line Search Newton with Modification

for $k = 0, 1, 2, \dots$

Factorize $B_k = \nabla^2 f_k + E_k$, with $E_k = 0$ if $\nabla^2 f_k$ is sufficiently positive definite, otherwise E_k chosen so that B_k is;

Solve $B_k p_k = -\nabla f_k$;

$x_{k+1} \leftarrow x_k + \alpha_k p_k$ with α_k Wolfe / Goldstein / backtracking.

Global convergence (Theorem 3.8): if the level set is compact and the *bounded modified factorization property* holds — $\kappa(B_k) = \|B_k\| \|B_k^{-1}\| \leq C$ whenever $\{\nabla^2 f_k\}$ is bounded (3.39) — then $\nabla f_k \rightarrow 0$. This is Zoutendijk again, via (3.19).

Rate: if $\nabla^2 f(x^*)$ is sufficiently positive definite, the modification returns $E_k = 0$ for large k , $\alpha_k = 1$ is accepted (Thm 3.6), and the rate is quadratic. If $\nabla^2 f(x^*)$ is near-singular, E_k may never vanish and the rate can be only linear.

Desiderata for E_k : keeps B_k well conditioned; **as small as possible**; cheap to compute.

Eigenvalue modification — and why the obvious idea is bad

Suppose $\nabla f_k = (1, -3, 2)^T$ and $\nabla^2 f_k = \text{diag}(10, 3, -1)$, indefinite. Pure Newton step: $p_k^N = (-0.1, 1, 2)^T$, and $\nabla f_k^T p_k^N > 0$ — *ascent*.

Obvious fix: replace the negative eigenvalue by a small $\delta > 0$, say $\delta = \sqrt{u} \approx 10^{-8}$:

$$B_k = \text{diag}(10, 3, 10^{-8}). \quad (3.41)$$

Numerically positive definite, curvature along q_1, q_2 preserved. But now

$$p_k = -B_k^{-1} \nabla f_k \approx -(2 \times 10^8) q_3. \quad (3.42)$$

Descent, but absurd

The step is essentially parallel to q_3 and enormously long. It violates the spirit of Newton's method, which trusts a quadratic model only *near* x_k .

Alternatives: flip signs of negative eigenvalues ($\delta = 1$ here); drop the negative-curvature component entirely; adapt δ to control step length (that is trust-region thinking, Ch. 4). *Great freedom, no agreement on what is best.*

Minimum-norm modifications

Given symmetric $A = Q\Lambda Q^T$, what is the *smallest* ΔA making $\lambda_{\min}(A + \Delta A) \geq \delta$?

Frobenius norm

$$\Delta A = Q \operatorname{diag}(\tau_i) Q^T, \quad \tau_i = \begin{cases} 0, & \lambda_i \geq \delta \\ \delta - \lambda_i, & \lambda_i < \delta \end{cases} \quad (3.43)$$

Not diagonal in general; touches only the offending eigenvalues (this is (3.41)).

Euclidean norm

$$\Delta A = \tau I, \quad \tau = \max(0, \delta - \lambda_{\min}(A)). \quad (3.44)$$

Shifts *all* eigenvalues. Modified matrix $A + \tau I$ (3.45) — **the same form as in trust-region methods** (Ch. 4).

Neither is used as written: the spectral decomposition is too expensive at scale. Practical schemes work *through Gaussian elimination*, choosing the modification indirectly. They often — not always — produce good steps.

Adding a multiple of the identity (Algorithm 3.3)

Find $\tau > 0$ such that $\nabla^2 f_k + \tau I$ is sufficiently positive definite, without knowing $\lambda_{\min}$: *try, and let Cholesky tell you.*

Algorithm 3.3 — Cholesky with Added Multiple of the Identity

Choose $\beta > 0$ (typically 10^{-3}).

If $\min_i a_{ii} > 0$ set $\tau_0 \leftarrow 0$; else $\tau_0 \leftarrow -\min_i a_{ii} + \beta$.

for $k = 0, 1, 2, \dots$

 Attempt Cholesky $LL^T = A + \tau_k I$;

 if it succeeds, return L ; else $\tau_{k+1} \leftarrow \max(2\tau_k, \beta)$.

- ▶ Simple, and often preferable to the fancier schemes below.
- ▶ **Cost:** every trial τ_k is a fresh factorization. If several trials are needed, grow τ by $10\times$ rather than $2\times$.
- ▶ Warm start: initialize τ_0 from the last successful τ .

What production codes actually do (keywords only)

Two families that modify the Hessian *during* factorization, with no trial-and-error:

- ▶ **Modified Cholesky** (Gill–Murray, Schnabel–Eskow). Run Cholesky; whenever a pivot is too small, increase it just enough. Guarantees the factors exist and stay bounded relative to $\|\nabla^2 f_k\|$; returns $E_k = 0$ when $\nabla^2 f_k$ is already sufficiently positive definite.
- ▶ **Modified symmetric indefinite factorization** (Bunch–Kaufman LBL^T). Factor with 1×1 and 2×2 pivots, then modify the small block-diagonal B by the minimum-Frobenius rule (3.43) blockwise. Handles indefiniteness more gracefully.

Why this matters beyond Chapter 3

The LBL^T factorization tells you the **inertia** of the matrix — how many positive, negative, and zero eigenvalues. In Chapter 19 the interior-point KKT system *must* have a specific inertia, and “inertia correction” is exactly τI added until it does. IPOPT’s log line `inertia correction` is Algorithm 3.3 in disguise.

Building the one-dimensional search

Task: find α minimizing $\phi(\alpha) = f(x_k + \alpha p_k)$ (3.54), or just satisfying a termination condition of §3.1. Assume $\phi'(0) < 0$.

- ▶ Convex quadratic $f = \frac{1}{2}x^T Qx - b^T x$: closed form $\alpha_k = -\nabla f_k^T p_k / (p_k^T Q p_k)$ (3.55). Otherwise iterate.
- ▶ Function-values-only procedures are inefficient: to be sound they must narrow the minimizer to a small interval. With *gradient* information the Wolfe/Goldstein conditions tell us immediately whether we are done — often on the very first trial.
- ▶ Structure: **bracketing** phase finds $[\bar{a}, \bar{b}]$ containing acceptable steps; **selection** phase zooms in, using interpolation of what has already been computed.

Notation: α_k, α_{k-1} are step lengths of the outer iteration; $\alpha_i, \alpha_{i-1}, \alpha_j$ are *trial* values inside one line search; α_0 the initial guess.

Interpolation: a smarter backtracking

Goal: satisfy sufficient decrease $\phi(\alpha) \leq \phi(0) + c_1 \alpha \phi'(0)$ (3.56) without α becoming “too small”. Use what we already know about ϕ .

Quadratic step. If α_0 fails, we know $\phi(0)$, $\phi'(0)$, $\phi(\alpha_0)$. Fit the quadratic through them,

$$\phi_q(\alpha) = \left(\frac{\phi(\alpha_0) - \phi(0) - \alpha_0 \phi'(0)}{\alpha_0^2} \right) \alpha^2 + \phi'(0) \alpha + \phi(0), \quad (3.57)$$

and take its minimizer

$$\alpha_1 = - \frac{\phi'(0) \alpha_0^2}{2[\phi(\alpha_0) - \phi(0) - \phi'(0) \alpha_0]}. \quad (3.58)$$

Cubic step. If α_1 also fails, fit a cubic through $\phi(0)$, $\phi'(0)$, $\phi(\alpha_0)$, $\phi(\alpha_1)$ and take its minimizer (in $[0, \alpha_1]$). Repeat with the two most recent values.

Safeguard. If α_i is too close to α_{i-1} or too much smaller, reset $\alpha_i = \alpha_{i-1}/2$. This is the “ $\rho \in [\rho_{lo}, \rho_{hi}]$ ” of Algorithm 3.1.

When derivatives are cheap (Ch. 8: directional derivative comes almost free with f), interpolate ϕ and ϕ' at the two most recent trials instead — (3.59). Cubic interpolation typically converges quadratically to the minimizer of ϕ .

The initial step length

Newton and quasi-Newton

$\alpha_0 = 1$, always. Otherwise the fast local rates of §3.3 are never observed.

Steepest descent, CG — directions with no natural scale

Use information from the previous iteration:

- ▶ first-order change in f equal to last step's, $\alpha_0 \nabla f_k^T p_k = \alpha_{k-1} \nabla f_{k-1}^T p_{k-1}$, i.e.
$$\alpha_0 = \alpha_{k-1} \nabla f_{k-1}^T p_{k-1} / \nabla f_k^T p_k;$$
- ▶ or the minimizer of the quadratic through f_{k-1} , f_k , $\nabla f_{k-1}^T p_{k-1}$: $\alpha_0 = 2(f_k - f_{k-1})/\phi'(0)$
(3.60)

If $x_k \rightarrow x^*$ superlinearly, the ratio in (3.60) tends to 1. Setting $\alpha_0 \leftarrow \min(1, 1.01\alpha_0)$ ensures the unit step is eventually tried and accepted — so even these methods recover superlinear behavior when it is available.

A line search for the strong Wolfe conditions (Algorithm 3.5)

Guaranteed to find α satisfying (3.7) for any $0 < c_1 < c_2 < 1$, given descent p and f bounded below along p . Stage 1 *increases* α until it finds an acceptable step or a bracket; stage 2 (zoom) shrinks the bracket.

Algorithm 3.5

Set $\alpha_0 \leftarrow 0$, choose $\alpha_{\max} > 0$, $\alpha_1 \in (0, \alpha_{\max})$; $i \leftarrow 1$.

repeat

 Evaluate $\phi(\alpha_i)$;

if $\phi(\alpha_i) > \phi(0) + c_1\alpha_i\phi'(0)$ **or** $[\phi(\alpha_i) \geq \phi(\alpha_{i-1})$ and $i > 1]$: $\alpha_* \leftarrow \text{zoom}(\alpha_{i-1}, \alpha_i)$, **stop**;

 Evaluate $\phi'(\alpha_i)$;

if $|\phi'(\alpha_i)| \leq -c_2\phi'(0)$: $\alpha_* \leftarrow \alpha_i$, **stop**;

if $\phi'(\alpha_i) \geq 0$: $\alpha_* \leftarrow \text{zoom}(\alpha_i, \alpha_{i-1})$, **stop**;

 Choose $\alpha_{i+1} \in (\alpha_i, \alpha_{\max})$; $i \leftarrow i + 1$.

The bracket (α_{i-1}, α_i) is known to contain a strong-Wolfe point if any of: (i) α_i violates sufficient decrease; (ii) $\phi(\alpha_i) \geq \phi(\alpha_{i-1})$; (iii) $\phi'(\alpha_i) \geq 0$. Trial values increase (extrapolate, or multiply by a constant) fast enough to hit $\alpha_{\max}$ in finitely many steps.

zoom (Algorithm 3.6)

Called as $\text{zoom}(\alpha_{\text{lo}}, \alpha_{\text{hi}})$ — note the argument order is *not* “low, high”. The invariants:

- (a) the interval between them contains strong-Wolfe step lengths;
- (b) α_{lo} has the *smallest* ϕ value among trials satisfying sufficient decrease;
- (c) $\phi'(\alpha_{\text{lo}})(\alpha_{\text{hi}} - \alpha_{\text{lo}}) < 0$ — ϕ decreases from α_{lo} toward α_{hi} .

Algorithm 3.6

repeat

Interpolate (quadratic, cubic, or bisection) a trial α_j between α_{lo} and α_{hi} ;

Evaluate $\phi(\alpha_j)$;

if $\phi(\alpha_j) > \phi(0) + c_1\alpha_j\phi'(0)$ **or** $\phi(\alpha_j) \geq \phi(\alpha_{\text{lo}})$: $\alpha_{\text{hi}} \leftarrow \alpha_j$;

else

Evaluate $\phi'(\alpha_j)$;

if $|\phi'(\alpha_j)| \leq -c_2\phi'(0)$: $\alpha_* \leftarrow \alpha_j$, **stop**;

if $\phi'(\alpha_j)(\alpha_{\text{hi}} - \alpha_{\text{lo}}) \geq 0$: $\alpha_{\text{hi}} \leftarrow \alpha_{\text{lo}}$;

$\alpha_{\text{lo}} \leftarrow \alpha_j$.

Each pass replaces one endpoint by α_j so that (a)–(c) persist. Interpolation must be safeguarded (keep α_j away from the endpoints) so the interval actually shrinks. This is what `scipy.optimize.line_search` and most quasi-Newton codes implement.

Chapter 3 in one table

Condition / method	Uses	Suited to
Armijo (3.4) alone	f	nothing — admits arbitrarily short steps
Armijo + backtracking (Alg. 3.1)	f	Newton
Wolfe (3.6)	f, ϕ'	everything; <i>required</i> for quasi-Newton
Strong Wolfe (3.7)	f, ϕ'	quasi-Newton, nonlinear CG
Goldstein (3.11)	f	Newton; not quasi-Newton
Zoutendijk (3.14)	—	global convergence for any $\cos \theta_k \geq \delta$
Theorem 3.3/3.4	—	steepest descent: linear, rate $\left(\frac{\kappa-1}{\kappa+1}\right)^2$
Theorem 3.5	—	Newton: quadratic, needs $\alpha = 1$
Theorem 3.6/3.7	—	quasi-Newton: superlinear iff Dennis–Moré (3.36)
Hessian modification (§3.4)	factorization	make Newton descend far from x^*
Algorithms 3.5/3.6	f, ϕ'	the strong-Wolfe line search in practice

Two ideas to carry forward

Each Wolfe condition has one job, and Zoutendijk's proof shows which. Fast local convergence is a property you can throw away by not trying $\alpha = 1$.

Homework

Derivations

- 1 Write out the proof of Lemma 3.1 in full, and indicate where each hypothesis is used.
- 2 Write out the proof of Theorem 3.2 (Zoutendijk) in full. Then explain, in one sentence each, what fails if (3.6a) is dropped, and what fails if (3.6b) is dropped.
- 3 (Exercise 3.5) Show that $\|B_k\| \|B_k^{-1}\| \leq M$ with $B_k \succ 0$ implies $\cos \theta_k \geq 1/M$.
- 4 (Exercise 3.7) Derive (3.28) for exact-line-search steepest descent on the quadratic (3.24).
- 5 Verify that the minimizer of (3.57) is (3.58), and that it lies in $(0, \alpha_0)$ when α_0 violates sufficient decrease.

Implementation

6. Implement backtracking (Alg. 3.1) and the strong-Wolfe search (Alg. 3.5–3.6). Run steepest descent and Newton on the Rosenbrock function from $x_0 = (-1.2, 1)$. Report iteration counts and the sequence of α_k ; observe when Newton starts taking $\alpha_k = 1$.
7. Implement Algorithm 3.3 and run Newton on a function whose Hessian is indefinite at x_0 (e.g. $f(x) = x_1^4 - x_1^2 + x_2^2$ from $(0.1, 1)$). Log τ_k per iteration.

Next: Ch. 4 — *distance first, direction second*.